Aditya Sharma

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Summary

Full Stack Software Engineer experienced in building end-to-end web applications with **React.js, Node.js, Express.js, PostgreSQL, and MongoDB**. Skilled across the stack: **responsive React UIs**, **REST API design**, **role-based access control**, authentication flows, and relational/non-relational database schemas. Engineered a **3-tier RBAC system across 40+ endpoints** with a matching React frontend, and a Tree-sitter-based static analysis engine with an interactive ReactFlow dependency graph that **cut LLM token usage by ∼98%**. Solved **400+ Data Structures & Algorithms problems** on LeetCode. Seeking a Full Stack Software Engineering Internship with the goal of converting to a full-time role.

Education

GITA Autonomous College — Bhubaneswar, India
B.Tech in Computer Science and Technology — ***CGPA: 9.54*** — *2024 – 2028*
- **Relevant Coursework:** Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Computer Networks

Projects

Requesta | *React, Express, MongoDB, Mongoose, OpenRouter API* — Live Demo | GitHub
- Engineered a **3-tier RBAC (Role-Based Access Control)** system across **40+ RESTful endpoints**, implementing a **backend workflow engine** to enforce approval-state transitions and prevent unauthorized actions
- Integrated the OpenRouter API across **4 AI-assisted workflows** — request drafting, quality scoring, approval suggestions, and system insights — with **automatic fallback** (GPT-4o mini → Claude 3.5 Sonnet) and **JSON-mode validation** for reliability
- Implemented comprehensive **audit logging** for every leave/certificate action and developed **analytics endpoints** for decision intelligence and approval trend analysis for administrators
- Secured user authentication using **dual-token JWT sessions** and **OTP-based password recovery**, balancing usability with credential protection for sensitive data
- Refactored **40+ frontend API calls** into reusable utility modules and automated transactional emails across **7 Resend templates**, decoupling UI components from network logic

RepoLens | *React, ReactFlow, Express, Tree-sitter, PostgreSQL, Prisma, OpenRouter API* — Live Demo | GitHub
- Architected a **full-stack code intelligence platform** integrating the GitHub API (OAuth, Octokit) to ingest repositories and execute a **multi-engine deterministic static analysis pipeline**, performing repository analysis **without exposing raw source code to third-party LLMs**
- Built a **Tree-sitter-based CST (Concrete Syntax Tree) extraction engine** to compute **cross-file dependency graphs**, LOC, nesting depth, and **cyclomatic/cognitive complexity**, and rendered results via an **interactive ReactFlow dependency graph**
- Engineered a **rule-based SAST engine** combining regex and CST tree-walking to flag unsafe patterns (`eval()`, SQL injection, path traversal, hardcoded secrets) with deterministic accuracy
- Optimized AI-assisted reporting by compressing analysis output into structured JSON before invoking the OpenRouter API, **reducing LLM token usage by ∼98%** and grounding the LLM in deterministic scan results to minimize hallucination
- Modeled a **6-table PostgreSQL schema** using Prisma and designed REST endpoints to persist scan history, run repository comparisons, and power AI-assisted chat over deterministic scan context

RagCoder Dark Theme | *VS Code Extension API, JSON, Marketplace* — Marketplace | GitHub
- Designed and published a custom VS Code color theme (Hyper-Vibrant Harmony palette) with token scope rules for **7+ programming language grammars**, ensuring consistent syntax highlighting across JavaScript, Python, Java, and C/C++
- Published and maintained the extension on the VS Code Marketplace using `vsce`, reaching **80+ installs** from external developers

Technical Skills

Frontend: React.js, React Router DOM, Tailwind CSS, Responsive Design, Framer Motion, GSAP, ReactFlow
Backend: Node.js, Express.js, REST API Design, JWT Authentication, RBAC (Role-Based Access Control)
Databases: PostgreSQL, MongoDB, Prisma, Mongoose, SQL, Database Design
Programming Languages: JavaScript, HTML, CSS
Libraries & APIs: Tree-sitter, OpenRouter API, GitHub API (Octokit), OAuth, Cloudinary, Resend
Developer Tools: Git, GitHub, Postman, VS Code Extension API
Cloud & Deployment: Vercel, Render
Core CS: Data Structures & Algorithms, Object-Oriented Programming (OOP), Database Management Systems

Achievements & Certifications

Solved **400+ Data Structures and Algorithms problems** on LeetCode, demonstrating strong problem-solving and algorithmic thinking
Programming with JavaScript — Meta (Coursera)
Published the RagCoder Dark Theme extension on the **VS Code Marketplace** with **80+ installs**